



Statistics of Financial Markets

Chapter 14: Fractal Market Hypothesis



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Outline

Motivation

Fractals in Finance

The Hurst Exponent

Investment Horizons

Empirical Evidence

Conclusions

References

Motivation

- The **Efficient Market Hypothesis** (EMH) of Fama (1970) has been the dominant paradigm in financial economics for over half a century
- EMH assumes that asset prices fully reflect all available information and that returns follow a random walk with i.i.d. Gaussian innovations:

$$r_t = \mu + \varepsilon_t, \quad \varepsilon_t \stackrel{\text{i.i.d.}}{\sim} \mathcal{N}(0, \sigma^2)$$

- However, EMH **fails to explain** several empirical regularities:
 - ▶ **Fat tails:** $\Pr(|r_t| > x) \gg \Phi(-x)$ for large x , i.e. return distributions exhibit excess kurtosis $\kappa > 3$
 - ▶ **Volatility clustering:** large returns tend to follow large returns, $\text{Corr}(|r_t|, |r_{t+k}|) > 0$ for many lags k
 - ▶ **Long memory:** slow decay of autocorrelations, $\rho(k) \sim Ck^{2H-2}$ as $k \rightarrow \infty$ for $H > 0.5$
 - ▶ **Market crashes:** sudden, correlated drops that are orders of magnitude beyond Gaussian predictions (e.g., Black Monday 1987: -22.6%)
- The **Fractal Market Hypothesis** (FMH), proposed by Peters (1991, 1994), offers an alternative framework based on fractal geometry, heterogeneous time horizons, and the concept of market liquidity

EMH vs FMH

Efficient Market Hypothesis (EMH)

- All investors have **the same investment horizon** and react identically to information
- Information is **immediately and fully** reflected in asset prices
- Returns are **i.i.d.** with a well-defined variance:

$$r_t \stackrel{\text{i.i.d.}}{\sim} F(\mu, \sigma^2)$$

- Markets are always in **equilibrium**; arbitrage opportunities vanish instantaneously
- Price changes follow a **random walk** $\succ$ no serial dependence
- Risk is fully captured by **variance** σ^2 (mean-variance framework)

Fractal Market Hypothesis (FMH)

- Investors operate on **multiple time horizons**: from intraday to multi-year
- Information is **valued differently** depending on the investor's horizon
- Market **stability** depends on the **diversity of horizons**:

$$\text{Liquidity} \propto \text{Horizon diversity}$$

- When horizons **collapse** (all investors panic simultaneously), liquidity evaporates and crashes occur
- Returns exhibit **self-similarity** and **long-range dependence**

- Risk is characterized by the **Hurst exponent** H and the full distributional shape

Self-Similarity

Definition 1 (Statistical Self-Similarity)

A stochastic process $\{X(t)\}_{t \geq 0}$ is called **self-similar** with index $H > 0$ (or H -ss) if for every $c > 0$:

$$\{X(ct)\}_{t \geq 0} \stackrel{d}{=} \{c^H X(t)\}_{t \geq 0}$$

where $\stackrel{d}{=}$ denotes equality of all finite-dimensional distributions.

- The parameter H is called the **self-similarity exponent** (or Hurst exponent)
- **Implications for returns:** if the log-price process $\ln P(t)$ is H -self-similar, then rescaling time by c is equivalent to rescaling returns by c^H :

$$r(t, \Delta) = \ln P(t + \Delta) - \ln P(t) \stackrel{d}{=} c^H r\left(\frac{t}{c}, \frac{\Delta}{c}\right)$$

- For returns over horizon Δ , the **volatility scaling law** follows:

$$\sigma(\Delta) = \sigma(1) \cdot \Delta^H$$

- Under EMH ($H = 0.5$), this gives the classical square-root-of-time rule: $\sigma(\Delta) = \sigma(1)\sqrt{\Delta}$
- For $H \neq 0.5$, the scaling deviates $\succ$ evidence **against** EMH

Fractional Brownian Motion

Definition 2 (Fractional Brownian Motion)

A **fractional Brownian motion** (fBm) $\{B_H(t)\}_{t \in \mathbb{R}}$ with Hurst parameter $H \in (0, 1)$ is the unique centered Gaussian process with $B_H(0) = 0$ and covariance:

$$\mathbb{E}[B_H(t) B_H(s)] = \frac{1}{2} (|t|^{2H} + |s|^{2H} - |t - s|^{2H})$$

□ Key properties of fBm:

- ▶ Self-similar with exponent H : $B_H(ct) \stackrel{d}{=} c^H B_H(t)$
- ▶ Stationary increments: $B_H(t + s) - B_H(s) \stackrel{d}{=} B_H(t)$
- ▶ Variance of increments: $\mathbb{E}[(B_H(t) - B_H(s))^2] = |t - s|^{2H}$

□ Three regimes based on H :

- ▶ $H = 0.5$: **Standard Brownian motion** $\succ$ independent increments, no memory
- ▶ $H > 0.5$: **Persistent** (trending) $\succ$ positive autocorrelation of increments,
 $\text{Corr}(\Delta B_H(t), \Delta B_H(t + k)) > 0$
- ▶ $H < 0.5$: **Anti-persistent** (mean-reverting) $\succ$ negative autocorrelation,
 $\text{Corr}(\Delta B_H(t), \Delta B_H(t + k)) < 0$

- The autocovariance of increments at lag k is:

Long Memory Processes

Definition 3 (ARFIMA Process)

A process $\{y_t\}$ follows an **ARFIMA**(p, d, q) model if:

$$\Phi(L)(1-L)^d y_t = \Theta(L)\varepsilon_t, \quad \varepsilon_t \stackrel{\text{i.i.d.}}{\sim} (0, \sigma_\varepsilon^2)$$

where $d \in (-0.5, 0.5)$ is the fractional differencing parameter, $\Phi(L) = 1 - \phi_1 L - \dots - \phi_p L^p$ is the AR polynomial, and $\Theta(L) = 1 + \theta_1 L + \dots + \theta_q L^q$ is the MA polynomial.

- The **fractional difference operator** $(1-L)^d$ is defined via its binomial expansion:

$$(1-L)^d = \sum_{k=0}^{\infty} \binom{d}{k} (-L)^k = 1 - dL - \frac{d(1-d)}{2!} L^2 - \frac{d(1-d)(2-d)}{3!} L^3 - \dots$$

- **Connection to the Hurst exponent:** for a pure ARFIMA(0, d , 0) process,

$$d = H - \frac{1}{2}$$

- The spectral density near frequency zero behaves as $f(\lambda) \sim C_f |\lambda|^{-2d}$ as $\lambda \rightarrow 0$

- **Long memory** ($0 < d < 0.5$, equivalently $H > 0.5$): the autocorrelation function decays hyperbolically, $\rho(k) \sim C k^{-2d-1}$ as $k \rightarrow \infty$, and $\sum_{k=-\infty}^{\infty} |\rho(k)| = \infty$

Hurst Exponent: Definition

- ▣ Named after **Harold Edwin Hurst** (1880–1978), a British hydrologist who studied the long-term storage capacity of the Nile River
- ▣ Hurst (1951) found that the annual minima and maxima of the Nile's water levels exhibited **long-range dependence**: wet years tended to cluster, as did dry years
- ▣ He discovered that the rescaled range R/S scaled as n^H with $H \approx 0.73$, far exceeding $H = 0.5$ expected for independent observations

Definition 4 (Hurst Exponent)

The **Hurst exponent** $H \in (0, 1)$ characterizes the long-range dependence structure of a time series. Its value determines the nature of the process:

- ▣ $H = 0.5$: **Random walk** (no memory) $\succ$ independent increments, consistent with EMH
- ▣ $H > 0.5$: **Persistent** / trending behavior $\succ$ positive long-range autocorrelation; past trends tend to continue
- ▣ $H < 0.5$: **Anti-persistent** / mean-reverting $\succ$ negative long-range autocorrelation; past trends tend to reverse
- ▣ For a self-similar process, the variance of aggregated returns over a window of size n scales as:

R/S Analysis

- The **Rescaled Range** (R/S) analysis is the classical method for estimating H (Hurst, 1951; Mandelbrot & Wallis, 1969)
- **Algorithm:** given a time series $\{x_1, x_2, \dots, x_N\}$:
 1. Divide the series into blocks of size n
 2. For each block $\{x_1, \dots, x_n\}$, compute the mean $\bar{x}_n = \frac{1}{n} \sum_{i=1}^n x_i$
 3. Compute the **cumulative deviations**: $Y_k = \sum_{i=1}^k (x_i - \bar{x}_n)$, $k = 1, \dots, n$
 4. Compute the **range**: $R(n) = \max_{1 \leq k \leq n} Y_k - \min_{1 \leq k \leq n} Y_k$
 5. Compute the **standard deviation**: $S(n) = \sqrt{\frac{1}{n} \sum_{i=1}^n (x_i - \bar{x}_n)^2}$
 6. Compute the **rescaled range**: $(R/S)_n = R(n)/S(n)$
- The **Hurst law**: as $n \rightarrow \infty$,

$$\mathbb{E} \left[\frac{R(n)}{S(n)} \right] \propto n^H$$

- Estimate H by **OLS regression** of $\log(R/S)$ on $\log(n)$:

$$\log \left(\frac{R(n)}{S(n)} \right) = \hat{H} \cdot \log(n) + c + \text{error}$$

- **Limitation:** R/S analysis is sensitive to short-range dependence and non-stationarity; Lo (1991) proposed a modified R/S statistic that accounts for short-range correlations

Detrended Fluctuation Analysis (DFA)

- **DFA** was introduced by Peng et al. (1994) as a robust alternative to R/S analysis, especially for non-stationary time series
- **Algorithm:** given a time series $\{x_1, \dots, x_N\}$:
 1. Compute the **integrated (profile) series**: $Y_k = \sum_{i=1}^k (x_i - \bar{x})$, $k = 1, \dots, N$
 2. Divide $\{Y_k\}$ into non-overlapping segments of length n
 3. In each segment, fit a polynomial trend $\hat{Y}_k^{(m)}$ of order m (DFA- m ; linear: $m = 1$, quadratic: $m = 2$)
 4. Compute the **fluctuation function**:

$$F(n) = \sqrt{\frac{1}{N} \sum_{k=1}^N \left(Y_k - \hat{Y}_k^{(m)} \right)^2}$$

5. The DFA scaling law gives: $F(n) \propto n^H$
- Estimate H from the slope of $\log F(n)$ vs. $\log n$
 - **Advantages over R/S analysis:**
 - ▶ Robust to **polynomial trends** (detrending removes non-stationarity artifacts)
 - ▶ Works for **non-stationary** series (integrated processes)
 - ▶ Higher-order DFA ($m \geq 2$) removes more complex trends
 - **Interpretation of DFA exponents:** $H < 0.5$: anti-persistent; $H = 0.5$: uncorrelated; $0.5 < H < 1$: long-range persistent; $H = 1$: $1/f$ noise; $H > 1$: non-stationary, unbounded

Multiple Investment Horizons

- The **central insight** of the Fractal Market Hypothesis: financial markets are populated by investors operating on **vastly different time scales**:
 - ▶ **High-frequency traders**: holding periods of milliseconds to minutes
 - ▶ **Day traders**: intraday to several days
 - ▶ **Swing/position traders**: weeks to months
 - ▶ **Mutual funds and hedge funds**: months to years
 - ▶ **Pension funds and sovereign wealth funds**: years to decades
- Each class of investor **values information differently** depending on their horizon:
 - ▶ A day trader reacts strongly to a disappointing earnings report
 - ▶ A pension fund ignores short-term noise and focuses on long-term fundamentals
- **Market stability** requires sufficient **diversity of horizons**. When short-horizon sellers panic, long-horizon investors step in to buy, providing liquidity:

Stable market $\iff$ Broad distribution of investment horizons

- The **fractal structure** of the market arises because the same price patterns appear at different time scales $\succ$ statistical self-similarity of returns
- Peters (1994) argues that a well-functioning market is like a fractal: **remove any one scale and the structure breaks down**

Liquidity and Market Crashes

- Under FMH, **liquidity** is not an abstract concept but a direct consequence of horizon diversity:

$$\text{Liquidity}_t \propto \sum_{h \in \mathcal{H}} w_h \cdot \text{Activity}_{h,t}$$

where $\mathcal{H}$ is the set of investment horizons and w_h are horizon-specific weights

- Normal market conditions:** investors at different horizons provide a **continuous flow** of buy and sell orders $\succ$ tight bid-ask spreads, deep order books
- Crisis mechanism** (FMH explanation of crashes):
 - A **fundamental shock** (e.g., Lehman collapse, COVID-19 pandemic) affects **all horizons simultaneously**
 - Long-horizon investors abandon their usual strategies and adopt **short-term behavior** (panic selling)
 - The distribution of horizons **collapses** $\succ$ all investors effectively have the same horizon
 - With no diversity, **liquidity evaporates**: there are no buyers for the sellers
 - The **fractal structure breaks down** $\succ$ the Hurst exponent shifts abruptly
- Empirical evidence:**
 - 2008 GFC: the Hurst exponent of the S&P 500 dropped sharply, indicating a loss of fractal structure (Dominique & Rivera, 2011)
 - COVID-19 (March 2020): unprecedented volatility and illiquidity as horizon diversity collapsed within

Hurst Exponent Estimates

- Extensive empirical literature documents departures from $H = 0.5$ across asset classes

Asset class	Market	$\hat{H}$ (typical)	Method
Stock indices	S&P 500	0.55–0.62	R/S, DFA
Stock indices	FTSE 100	0.53–0.60	DFA
Stock indices	DAX	0.54–0.61	R/S, DFA
Emerging markets	BIST 100, BET	0.58–0.68	DFA
Cryptocurrencies	Bitcoin	0.50–0.75	DFA, wavelet
Cryptocurrencies	Ethereum	0.52–0.70	DFA
Forex	EUR/USD	0.50–0.55	R/S, DFA
Commodities	Gold	0.52–0.58	DFA
Bonds	US 10Y Treasury	0.55–0.65	R/S

- Key findings:** most equity indices exhibit mild persistence ($H \approx 0.55\text{--}0.65$), consistent with FMH
- Cryptocurrencies show **highly variable** H , reflecting evolving market microstructure and varying degrees of efficiency

Multifractal Models

- A single Hurst exponent H (monofractal) may be insufficient to capture the full complexity of financial returns
- Mandelbrot, Fisher, and Calvet (1997) introduced the **Multifractal Model of Asset Returns** (MMAR)
- In the MMAR, the log-price is modeled as:

$$\ln P(t) = B_H[\theta(t)]$$

where B_H is a fractional Brownian motion and $\theta(t)$ is a **multifractal trading time** (a random measure)

- The process is **multifractal** if its q -th absolute moments scale as:

$$\mathbb{E}[|X(t)|^q] = c(q) \cdot t^{\tau(q)+1}$$

where $\tau(q)$ is a **nonlinear** function of q (for monofractal processes, $\tau(q) = qH - 1$ is linear)

- The **multifractal spectrum** $f(\alpha)$ describes the distribution of local Hölder exponents α :

$$f(\alpha) = \inf_q \{q\alpha - \tau(q)\} \quad (\text{Legendre transform})$$

- A **wide** $f(\alpha)$ spectrum indicates strong multifractality $\succ$ the series has regions of very different local scaling behavior
- Advantages over monofractal models: captures **volatility clustering**, **fat tails**, and **intermittency** simultaneously

Applications

- **Estimating the Hurst exponent in Python:** use the `nolds` or `hurst` libraries for R/S and DFA:

```
import numpy as np
import nolds

# Simulated fractional Brownian motion (H = 0.7)
from fbm import FBM
f = FBM(n=10000, hurst=0.7, length=1, method='daviesharte')
fbm_sample = f.fbm()
increments = np.diff(fbm_sample)

# R/S estimate
H_rs = nolds.hurst_rs(increments)
print(f"R/S_Hurst_estimate: {H_rs:.4f}")

# DFA estimate
H_dfa = nolds.dfa(increments)
print(f"DFA_Hurst_estimate: {H_dfa:.4f}")
```

- **Rolling Hurst exponent:** compute $\hat{H}_t$ in a rolling window of size w to detect **regime changes**:

$$\hat{H}_t = \text{DFA}(\{r_{t-w+1}, \dots, r_t\}), \quad t = w, w+1, \dots, T$$

- A **sudden drop** in $\hat{H}_t$ towards 0.5 (or below) signals loss of fractal structure $\succ$ potential crisis indicator
- **Practical application:** FMH-based risk management monitors horizon diversity via time-varying H_t estimates as an **early warning signal** for market stress

Conclusions

- The **Fractal Market Hypothesis** provides a richer framework than EMH for understanding financial markets:
 - ▶ Markets are stable when populated by investors with **diverse time horizons**
 - ▶ Crashes occur when this diversity **collapses** and all investors act on the same short-term horizon
 - ▶ Returns exhibit **self-similarity** and **long-range dependence**, captured by the Hurst exponent H
- **Comparison with EMH:**
 - ▶ EMH: $H = 0.5$, i.i.d. returns, no crashes beyond Gaussian tails
 - ▶ FMH: $H \neq 0.5$, fractal scaling, horizon-dependent liquidity, structural explanation for crashes
- **Practical implications for risk management:**
 - ▶ Standard VaR models (assuming $H = 0.5$) **underestimate tail risk** when $H > 0.5$
 - ▶ Volatility scaling: $\sigma(\Delta) = \sigma(1) \cdot \Delta^H$ instead of $\sigma(1) \cdot \Delta^{0.5}$
 - ▶ Monitoring the time-varying Hurst exponent $\hat{H}_t$ as an **early warning indicator** for liquidity crises
 - ▶ Multifractal models (MMAR) offer improved density forecasts for portfolio optimization
- **Open research directions:** connection between FMH and adaptive market hypothesis (Lo, 2004), high-frequency fractal analysis, multifractal portfolio theory

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